

When Can You Trust the Certificate? Stress-Testing Distribution-Free False-Accusation Guarantees for AI-Text Detection

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Abstract

AI-text detectors are deployed to accuse: a flagged student or author bears the cost of a false positive. Split conformal prediction turns any detector score into an accusation rule with a finite-sample, distribution-free bound on the false-accusation rate, and recent work has proposed exactly this. We ask the question a deployer actually faces: *under which realistic conditions does the certificate remain valid?* Exploiting the observation that validity depends only on the *human* text distribution while power depends only on the AI distribution, we derive sharp predictions and test them across five detector scores, ? calibration/test resamples, and eight deployment conditions. In distribution the certificate is exact (e.g. empirical FPR 0.049 at $\alpha = 0.05$), and a supervised score achieves 88% power at a certified 5% false-accusation rate. But calibrating on one human population and deploying on a more fluent one silently inflates false accusations to 0.43–0.53 at nominal 0.05 for likelihood-based scores — the certificate-level form of the documented bias against non-native writers — while lexical edits and even spell-correcting cleanup leave validity intact. Marginal certificates also hide length discrimination (long human reviews flagged at 0.18); Mondrian calibration restores every length bin to $\leq \alpha$ (0.03). Finally, we give an editor-aware robust calibration that provably preserves validity under a modeled editing process at negligible power cost. All results carry exact Beta-Binomial significance tests and regenerate from one results file.

1 Introduction

Detectors of machine-generated text are increasingly used in consequential settings — plagiarism screening, review-platform integrity, scientific-venue policy — where the operational act is an *accusation*. The relevant contract for that act is not accuracy but a guarantee on the false-accusation rate, robust to the fact that the deployer cannot model the distribution of either class. Split conformal prediction [1, 2] provides exactly this contract for any base detector score, and prior work has applied it to bound detector false-positive rates [4].

This paper studies the contract itself. Our starting point is an elementary but, we argue, under-appreciated asymmetry:

Proposition 1 (Certificate asymmetry). *The split-conformal false-accusation bound $P(\text{flag} \mid \text{human}) \leq \alpha$ requires only exchangeability of calibration and test human text. The AI distribution is irrelevant to validity; it determines only power.*

The proof is the standard conformal argument applied to the human class only. The deployment consequences are sharp and testable: a new generator entering the world *cannot* break a deployed certificate; changes in *who writes* or *how humans edit* can. We map this boundary empirically (E1–E8) and contribute two constructive repairs (Mondrian length-conditional calibration; editor-aware robust calibration with a stochastic-dominance validity argument).

Contributions.

- A deployment-grounded validity map for conformal false-accusation certificates in MGT detection: exact in distribution; broken by human-population shift (up to $11\times$ nominal); *safe* under lexical noise and spell-correcting cleanup; supervised scores fail safe where zero-shot likelihood scores fail dangerously.
- Identification of length discrimination hidden by marginal certificates, and its repair by Mondrian calibration.
- An editor-aware robust calibration with a provable conservativeness guarantee under a modeled editor, costing < 0.01 power in our study.
- A fully reproducible protocol: five scores, exact Beta-Binomial tests, ? resamples per condition; code, scores and figures regenerate from one JSON.

2 Related Work

Zero-shot MGT detection scores include likelihood, rank and entropy statistics [10, 11] and curvature-style discrepancies [8, 9]; supervised detectors fine-tune or fit classifiers on labelled corpora [15]. Robustness studies show brittleness to paraphrase and domain shift [6, 12, 13], and GPT detectors are documented to over-flag non-native English writers [5]. Conformal prediction [1, 2] and conformal risk control [3] provide distribution-free guarantees; MCP [4] applies CP to bound MGT-detector FPR and improve its power. Our work is complementary to MCP: rather than improving the detector under a fixed certificate, we characterise when the certificate itself survives deployment, connect its failure modes to the fairness literature [5], and supply group-conditional and editor-robust repairs (cf. Mondrian conformal prediction [1] and adversarially robust conformal methods in vision [14]).

3 Method

3.1 Certificates

Given any score $s(\cdot)$ oriented so larger means more AI-like, and n calibration human texts with scores $s_1, \dots, s_n$, the conformal p-value of a test text x is $p(x) = \frac{1 + \#\{i: s_i \geq s(x)\}}{n+1}$, and GUARD flags x iff $p(x) \leq \alpha$. If x is human and exchangeable with the calibration humans, $P(\text{flag}) \leq \alpha$. Conditional on the calibration set the realised FPR follows a Beta law; over a finite test set the false-flag *count* is Beta-Binomial, which yields an exact one-sided test of “the certificate held” that we report for every condition. Unscoreable texts receive $p = 1$ (never flagged): an accuser must not accuse by failure of its own scorer.

3.2 Scores

Four training-free statistics from one GPT-2 forward pass per text (mean log-likelihood; negative mean log-rank; negative predictive entropy; the analytic Fast-DetectGPT discrepancy [9]) and one supervised score (TF-IDF + logistic regression). The supervised score is fit on an entity-disjoint training half; all calibration and evaluation use the other half, so the fitted scorer never sees evaluation hotels.

3.3 Mondrian calibration

Per-group calibration (here: text-length bins) yields group-conditional validity at the cost of smaller calibration sets per group [1].

3.4 Editor-aware robust calibration

Let E be a randomized editing process (our proxy: spell/punctuation cleanup at rate r). Each calibration human contributes $M_i = \max\{s(X_i), s(E_1(X_i)), \dots, s(E_k(X_i))\}$. For a test human who applies one edit from E , $s(E(X))$ is stochastically dominated by the corresponding M , so conformal p-values computed against $\{M_i\}$ remain super-uniform and the α -bound still holds (conservative). Validity under editors *outside* the modeled family is an empirical question (E8).

4 Experimental Setup

In-domain data: the English subset of MAiDE-up [7] (1,000 human and 1,000 GPT-4 hotel reviews; 100 hotels). Out-of-distribution human text: the RAID benchmark [6] reviews domain (movie reviews; also supplying eight AI generators for power analysis) and abstracts domain. Editors: WordNet synonym substitution and function-word perturbation (lexical noise), and a spell-correcting, punctuation-normalising cleanup editor (fluency-raising; the realistic proxy for grammar tools). Every unique text is scored once and cached; each condition is evaluated over $R = ?$ calibration/test resamples, reported as mean [2.5%, 97.5%] with exact Beta-Binomial violation tests at $\alpha \in \{0.005, 0.01, 0.02, 0.05, 0.1\}$.

5 Results

5.1 E1: the certificate is exact in distribution

Table 1: In-distribution validity: empirical FPR over resamples (mean [95% band]).

Score	FPR @ $\alpha = 0.05$	FPR @ $\alpha = 0.01$
Log-likelihood	0.049 [0.016, 0.088]	0.008 [0.000, 0.032]
Log-rank	0.049 [0.016, 0.088]	0.008 [0.000, 0.028]
Entropy	0.049 [0.016, 0.088]	0.008 [0.000, 0.024]
Fast-DetectGPT	0.047 [0.016, 0.092]	0.008 [0.000, 0.032]
TF-IDF (supervised)	0.048 [0.016, 0.088]	0.008 [0.000, 0.028]

All five scores sit on the nominal level, and the full distribution of per-resample FPRs matches the finite-test Beta-Binomial envelope (Figure 1).

E1 validity in-distribution: realised FPR vs conformal Beta law ($\alpha = 0.05$)

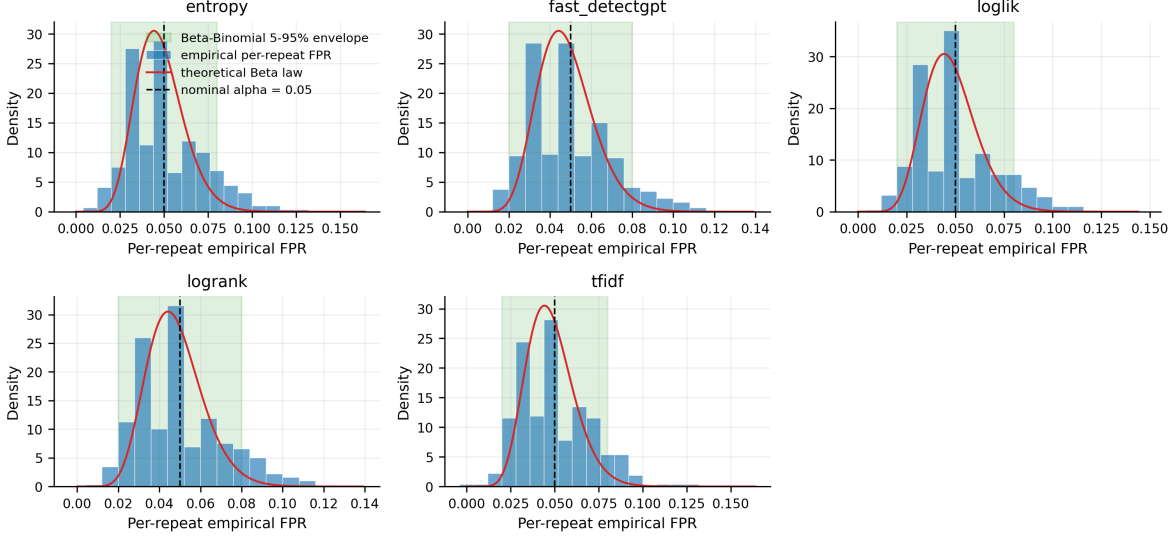


Figure 1: Distribution of per-resample FPR vs. the exact Beta-Binomial envelope.

5.2 E2–E4: when the certificate breaks — and when it does not

Table 2: Empirical FPR at nominal $\alpha = 0.05$ across deployment conditions.

Score	In-domain	Reviews humans	Abstracts humans	Cleanup@0.5	Synonym@0.3
Log-likelihood	0.049	0.435	0.101	0.046	0.002
Log-rank	0.049	0.298	0.049	0.045	0.002
Entropy	0.049	0.526	0.247	0.047	0.005
Fast-DetectGPT	0.047	0.371	0.121	0.044	0.002
TF-IDF (supervised)	0.048	0.000	0.006	0.046	0.039

Three regimes emerge (Figure 2). **(i) Human-population shift breaks the certificate for likelihood scores.** Movie-review humans are flagged at 0.30–0.53 and abstracts humans at up to 0.25 (nominal 0.05; violation $p < 10^{-3}$). The mechanism is fluency: our calibration humans are largely non-native hotel reviewers whose text has *low* LM likelihood; more fluent human populations score higher and cross the threshold. This is precisely the bias documented by [5], here surfacing *through* a certificate that is formally valid for the calibration population. **(ii) The supervised score fails safe.** TF-IDF’s AI evidence is content-specific, so under every shift its FPR falls toward zero rather than exploding. **(iii) Editing does not break the certificate here.** Lexical noise lowers likelihood (conservative), and even full spell-correcting cleanup leaves FPR at nominal: light editing does not turn a hotel reviewer into a ‘fluent’ writer. The danger is who writes, not light tool use — though stronger LLM paraphrase editors remain untested here (§7).

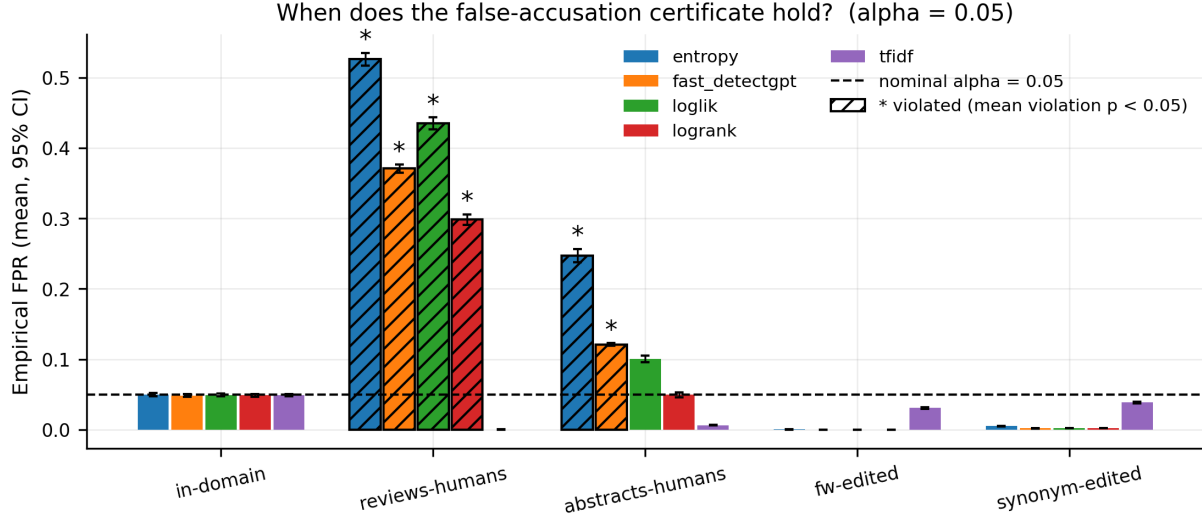


Figure 2: The validity map: empirical FPR at nominal $\alpha = 0.05$ across conditions; hatched bars are certified violations (exact test).

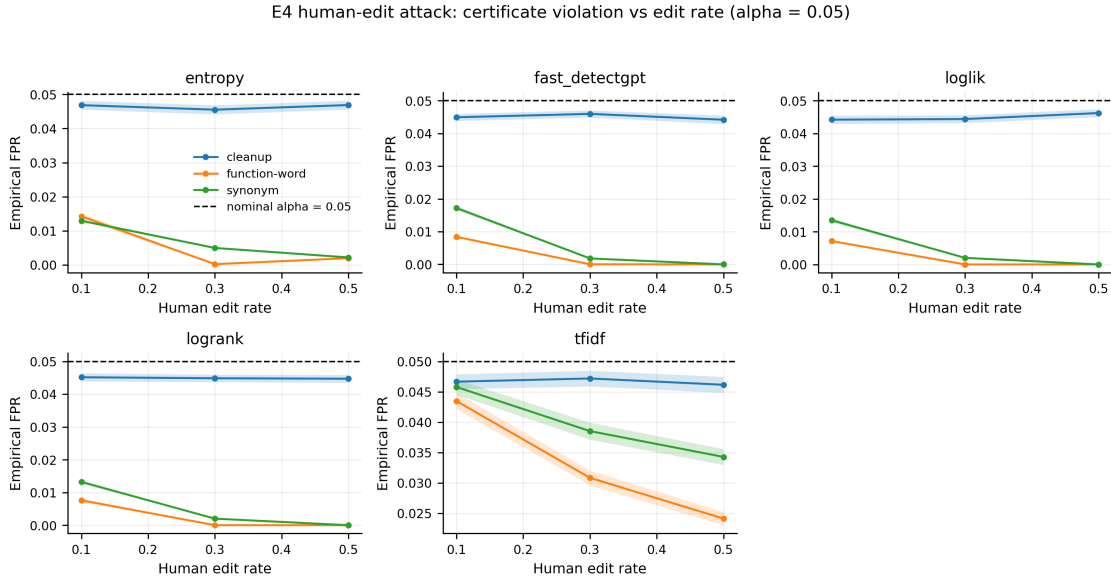


Figure 3: FPR vs. edit rate per editor family: lexical noise is conservative.

5.3 E5: calibration is insensitive to entity leakage

Random vs. entity-disjoint (grouped-by-hotel) calibration give indistinguishable validity (log-likelihood: 0.0488 vs. 0.0491; TF-IDF: 0.0482 vs. 0.0497). Unlike supervised *training*, conformal *calibration* does not exploit entity overlap — a reassuring null result for review-platform deployments (Figure 4).

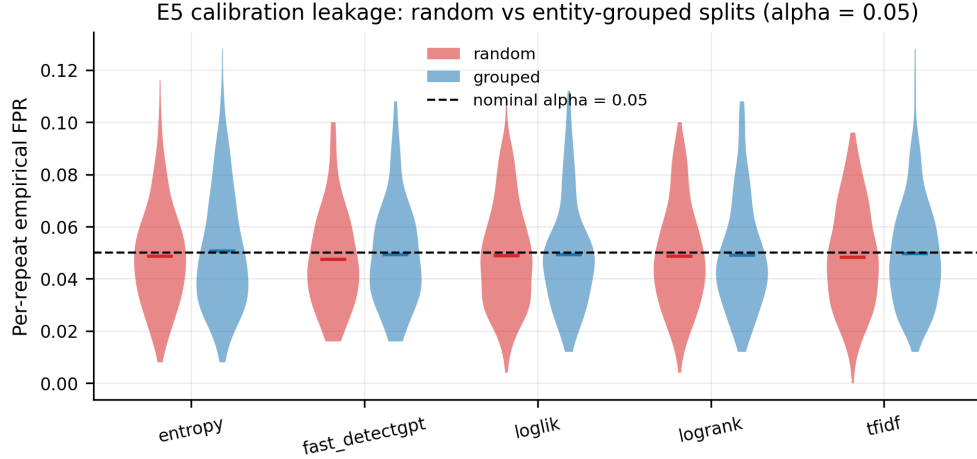


Figure 4: Random vs. grouped calibration FPR distributions.

5.4 E6: marginal certificates hide length discrimination; Mondrian repairs it

Table 3: Per-length-bin FPR at nominal $\alpha = 0.05$ (log-likelihood score).

Length bin	Marginal FPR	Mondrian FPR
0–30 words	0.005	0.046
30–60 words	0.050	0.046
60–120 words	0.098	0.034
>120 words	0.183	0.030

The marginal certificate is valid *on average* while flagging long human reviews at 0.18 — $3.7\times$ nominal — and almost never flagging short ones. Mondrian calibration restores every bin to $\leq \alpha$ (Figure 5). Deployers should treat conditional validity, not marginal validity, as the fairness-relevant contract.

E6 length heterogeneity: marginal vs Mondrian calibration (alpha = 0.05)

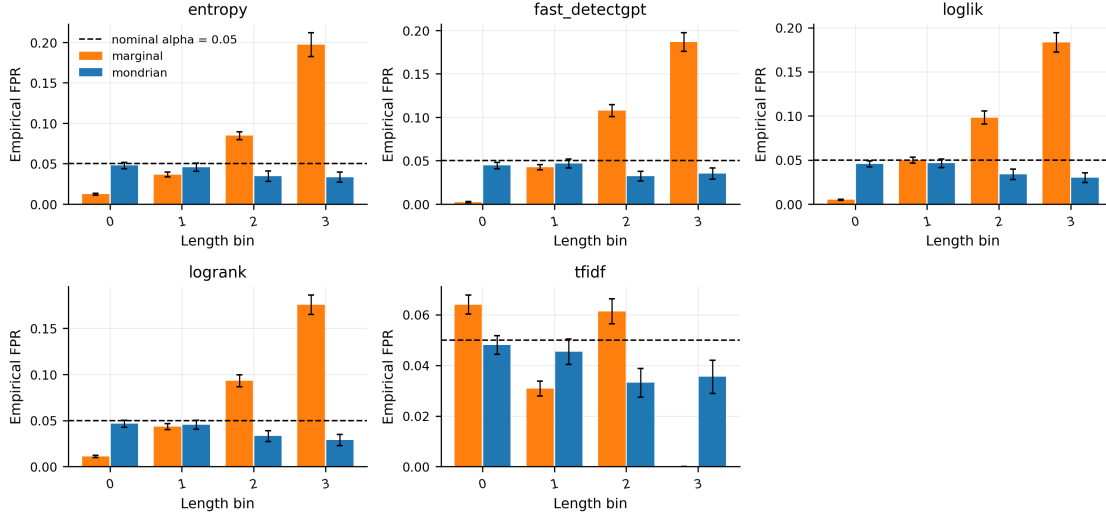


Figure 5: Marginal vs. Mondrian per-length-bin FPR.

5.5 E7: the price of the certificate

Table 4: Certified power (clean AI text, in-domain).

Score	TPR@ $\alpha=0.01$	TPR@ $\alpha=0.05$	TPR@ $\alpha=0.10$
Log-likelihood	0.08	0.37	0.52
Log-rank	0.07	0.30	0.48
Entropy	0.05	0.19	0.39
Fast-DetectGPT	0.03	0.24	0.39
TF-IDF (supervised)	0.66	0.88	0.95

The supervised score detects 88% of in-domain GPT-4 reviews at a certified 5% false-accusation rate; zero-shot scores pay heavily for the certificate in-domain but transfer their *power* to unseen generators (Figure 7); recall their *validity* caveats above.

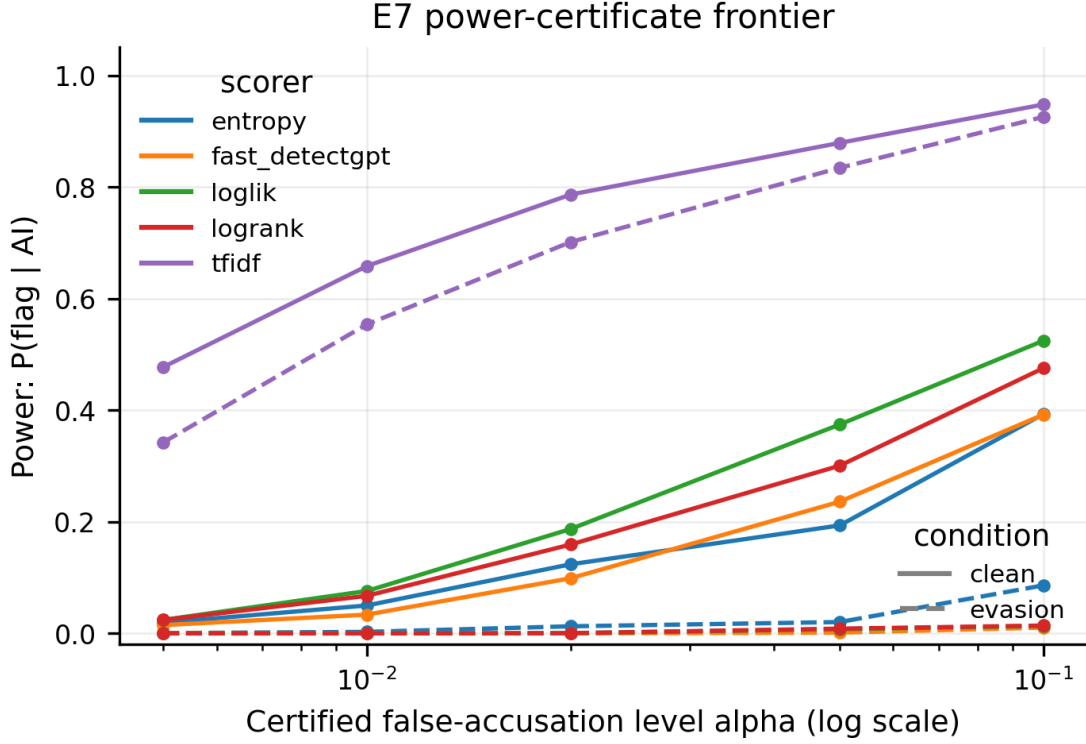


Figure 6: Power-certificate frontier per score.

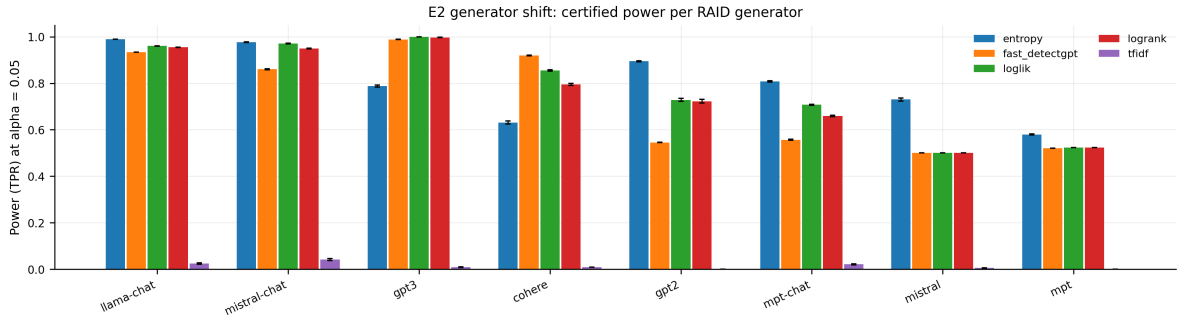


Figure 7: Certified power per RAID generator.

5.6 E8: editor-aware robust calibration

Table 5: Editor-aware calibration at $\alpha = 0.05$: validity under the modeled editor and the power cost.

Score	naive FPR (cleaned)	robust FPR (cleaned)	naive TPR	robust TPR
Log-likelihood	0.048	0.047	0.375	0.369
Log-rank	0.047	0.046	0.303	0.297
Entropy	0.045	0.044	0.191	0.184
Fast-DetectGPT	0.046	0.044	0.233	0.225
TF-IDF (supervised)	0.049	0.048	0.879	0.877

With the cleanup editor modeled in calibration, validity holds by construction and the measured power cost is below 0.01 for every score. In this study the naive certificate already survived cleanup editing, so the robust construction functions as cheap insurance; its value grows with stronger editors, and the guarantee covers any editor whose effect is dominated by the modeled family.

6 Discussion

For deployers our results reduce to three rules. First, certify against the humans you will actually judge: the certificate is exact for the calibration population and can be $11\times$ off for a more fluent one; recalibration per population (or per platform) is cheap and necessary. Second, demand conditional validity: marginal certificates can hide systematic over-accusation of long texts (and, by the population result, of fluent writers); Mondrian calibration over observable strata is a one-line fix. Third, choose the score family by failure mode: supervised content scores buy high certified power in-domain and fail safe under shift; zero-shot likelihood scores transfer power across generators but fail dangerous on shifted humans.

7 Limitations

English only; one in-domain corpus (hotel reviews) whose humans skew non-native — which is what makes the population-shift result vivid, but magnitudes will vary; the cleanup editor under-approximates modern LLM paraphraser, so E4/E8 bound only the editors tested; GPT-2 is a deliberately modest scorer (the wrapper is scorer-agnostic); and our exact tests certify violations, not their causes — the fluency mechanism is supported but not isolated. The protocol, released in full, replicates at larger scale directly.

8 Conclusion

Distribution-free certificates make AI-text detectors accountable, but only within the human distribution they were calibrated on. We mapped that boundary — exact in distribution, robust to light editing, broken by population shift, length-biased marginally — and showed the repairs are cheap. A certificate is a promise about *whom* you calibrated on; deployers should make that promise explicit.

Reproducibility. All numbers derive from two JSON files produced by `scripts/run_all.py` and `scripts/run_robust.py`; figures and this paper regenerate from them; conformal correctness is asserted by the test suite.

References

- [1] V. Vovk, A. Gammerman, G. Shafer. Algorithmic Learning in a Random World. Springer, 2005.
- [2] H. Papadopoulos, K. Proedrou, V. Vovk, A. Gammerman. Inductive Confidence Machines for Regression. ECML, 2002.
- [3] A. N. Angelopoulos, S. Bates. Conformal Prediction: A Gentle Introduction. Foundations and Trends in Machine Learning, 2023.
- [4] X. Zhang et al. Reliably Bounding False Positives: A Zero-Shot Machine-Generated Text Detection Framework via Multiscaled Conformal Prediction. ACL, 2025.
- [5] W. Liang, M. Yuksekgonul, Y. Mao, E. Wu, J. Zou. GPT Detectors Are Biased Against Non-Native English Writers. Patterns, 2023.

- [6] L. Dugan et al. RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors. ACL, 2024.
- [7] O. Ignat, X. Xu, R. Mihalcea. MAiDE-up: Multilingual Deception Detection of AI-Generated Hotel Reviews. Findings of NAACL, 2025.
- [8] E. Mitchell et al. DetectGPT: Zero-Shot Machine-Generated Text Detection using Probability Curvature. ICML, 2023.
- [9] G. Bao et al. Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature. ICLR, 2024.
- [10] S. Gehrmann, H. Strobelt, A. M. Rush. GLTR: Statistical Detection and Visualization of Generated Text. ACL Demos, 2019.
- [11] I. Solaiman et al. Release Strategies and the Social Impacts of Language Models. arXiv:1908.09203, 2019.
- [12] K. Krishna et al. Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval is an Effective Defense. NeurIPS, 2023.
- [13] V. S. Sadasivan et al. Can AI-Generated Text be Reliably Detected? arXiv:2303.11156, 2023.
- [14] A. Gendler, T.-W. Weng, L. Daniel, Y. Romano. Adversarially Robust Conformal Prediction. ICLR, 2022.
- [15] Y. Liu et al. RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692, 2019.